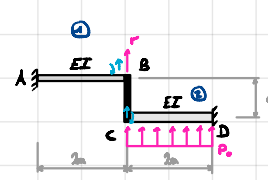


Problem 1) a) Find $\underline{K} = \underline{\tilde{K}}$

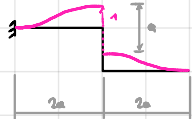


S1)

$$\underline{\tilde{v}} = \begin{bmatrix} v_{B1} \\ v_{B2} \\ v_{C1} \\ v_{C2} \end{bmatrix}$$

$$\underline{\tilde{v}} = \begin{bmatrix} v_{D1} \\ v_{D2} \\ v_{C1} \\ v_{C2} \end{bmatrix}$$

→ r = 1



$$\underline{\tilde{v}} = \underline{a} \underline{\tilde{r}} \quad \underline{\tilde{r}} = [r]$$

$$\Rightarrow \underline{a}^{(1)} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \underline{a}^{(2)} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

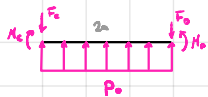
S3)

$$\underline{K} = \sum_{e=1}^N (\underline{a}^{(e)T}) \underline{K}^{(e)} \underline{a}^{(e)} = \frac{3EI}{a^3} \cdot \underline{\tilde{K}}$$

S4) consistent element force vector

$$\underline{\tilde{R}} = \underline{\tilde{R}}^{(e)} = \underline{\tilde{R}}^{(e)}$$

$$K_1 = K_2 = \frac{P_0^2}{12} \cdot \frac{P_0 a^2}{3} \quad F_1 = F_2 = \frac{P_0 L}{2} = P_0 a$$



$$\underline{\tilde{S}}^{(e)} = \begin{bmatrix} -\frac{P_0 a}{3} \\ \frac{P_0 a^2}{6} \\ -\frac{P_0 a}{3} \\ \frac{P_0 a^2}{6} \end{bmatrix} \quad \underline{\tilde{S}}^{(e)} = 0$$

$$\Rightarrow \underline{\tilde{R}}^{(e)} = \underline{a}^{(e)T} \underline{\tilde{S}}^{(e)} = [1 \ 0 \ 0 \ 0] \begin{bmatrix} -\frac{P_0 a}{3} \\ \frac{P_0 a^2}{6} \\ -\frac{P_0 a}{3} \\ \frac{P_0 a^2}{6} \end{bmatrix} = -\frac{P_0 a}{3}$$

$$\Rightarrow \underline{\tilde{R}} = -\underline{\tilde{R}}^{(e)} = P_0 a$$

S2)

$$\underline{\tilde{S}} = \underline{\tilde{K}} \underline{\tilde{v}}$$

$$\underline{a}^T \underline{K} \underline{a}$$

ELEMENT STIFFNESSES

$$\underline{AB}: [0 \ 0 \ 1 \ 0] \cdot \frac{EI}{L^3} \begin{bmatrix} 12 & -6L & -12 & -6L \\ -6L & 4L^2 & 6L & 2L^2 \\ -12 & 6L & 12 & 6L \\ -6L & 2L^2 & 6L & 4L^2 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -\frac{12EI}{L^3} \\ \frac{6EI}{L^3} \\ \frac{12EI}{L^3} \\ \frac{6EI}{L^3} \end{bmatrix}^T \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \frac{12EI}{L^3} = \frac{12EI}{8a^3} = \frac{3EI}{2a^3} = \underline{\tilde{K}}^{(1)}$$

$$\underline{CD}: [1 \ 0 \ 0 \ 0] \cdot \frac{EI}{L^3} \begin{bmatrix} 12 & -6L & -12 & -6L \\ -6L & 4L^2 & 6L & 2L^2 \\ -12 & 6L & 12 & 6L \\ -6L & 2L^2 & 6L & 4L^2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

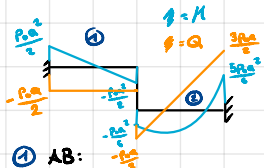
$$= \frac{12EI}{L^3} = \frac{3EI}{2a^3} = \underline{\tilde{K}}^{(2)}$$

S5)

$$\underline{\tilde{R}} = \underline{\tilde{K}} \underline{\tilde{r}} \Rightarrow P_0 a = \frac{3EI}{a^3} \cdot r$$

$$\Rightarrow r = \frac{P_0 a^4}{3EI}$$

b) moment & shear force distribution



$$r = \frac{P_0 a^4}{3EI}$$

① AB:

$$EI v'''' = 0$$

$$EI v''' = C_1$$

$$M(x) = EI v'' = C_1 x + C_2$$

$$EI v' = \frac{1}{2} C_1 x^2 + C_2 x + C_3$$

$$EI v = \frac{1}{6} C_1 x^3 + \frac{1}{2} C_2 x^2 + C_3 x + C_4$$

$$v(0) = 0 \Rightarrow C_4 = 0$$

$$v'(0) = 0 \Rightarrow C_3 = 0$$

$$v'(L) = 0 \Rightarrow \frac{1}{2} C_1 L^2 + C_2 L = 0 \Rightarrow C_2 = -\frac{1}{2} C_1 L$$

$$v(L) = 0 \Rightarrow \frac{1}{6} C_1 L^3 + \frac{1}{2} C_2 L^2 + C_3 L = 0 \Rightarrow C_1 = -\frac{12EI}{L^3} r$$

$$C_2 = \frac{6EI}{L^2} r$$

$$\Rightarrow M(x) = -\frac{12EI}{8a^3} x + \frac{6EI}{4a^2} = \frac{6EI}{4a^2} \left(1 - \frac{x}{a}\right)$$

$$x=0: \frac{P_0 a^2}{2} \quad x=2a: \frac{P_0 a^2}{2} (1-2) = -\frac{P_0 a^2}{2}$$

$$= \frac{P_0 a^2}{2} \left(1 - \frac{x}{a}\right)$$

P_0 only

② CD:

$$EI v'''' = P_0$$

$$EI v''' = P_0 x + C_1$$

$$M(x) = EI v'' = \frac{P_0}{2} x^2 + C_1 x + C_2$$

$$EI v' = \frac{P_0}{6} x^3 + \frac{C_1}{2} x^2 + C_2 x + C_3$$

$$EI v = \frac{P_0}{24} x^4 + \frac{C_1}{6} x^3 + \frac{C_2}{2} x^2 + C_3 x + C_4$$

$$v(0) = 0 \Rightarrow C_4 = 0$$

$$v'(0) = 0 \Rightarrow C_3 = 0$$

$$v'(L) = 0 \Rightarrow \frac{P_0}{6} L^3 + \frac{C_1}{2} L^2 + C_2 L = 0 \Rightarrow C_1 = -\frac{P_0}{6} L^3 - \frac{C_2}{2} L^2 \Rightarrow C_2 = -\frac{P_0}{6} L^3 + \frac{P_0}{4} L^2 = \frac{P_0}{12} L^2$$

$$v(L) = 0 \Rightarrow \frac{P_0}{24} L^4 + \frac{C_1}{6} L^3 + \frac{C_2}{2} L^2 = 0 \Rightarrow \frac{P_0}{24} L^4 + \frac{1}{6} \left(-\frac{P_0}{6} L^3 - \frac{C_2}{2} L^2\right) L^3 + \frac{P_0}{12} L^2 L^2 = 0$$

$$C_1 \left(\frac{1}{6} - \frac{1}{6}\right) - \frac{P_0}{24} L + \frac{P_0}{12} L = \frac{P_0}{24} L$$

$$C_1 = -\frac{P_0}{2} L$$

$$\Rightarrow M(x) = P_0 \left(\frac{x^2}{2} - \frac{Lx}{2} + \frac{L^2}{12}\right) = P_0 \left(\frac{x^2}{2} - ax + \frac{a^2}{3}\right)$$

disp r only

$$M(x) = -\frac{P_0 a^2}{2} \left(1 - \frac{x}{a}\right)$$

$$M(x) = M(x) + M_r(x) = P_0 \left(\frac{x^2}{2} - ax + \frac{a^2}{3}\right) - \frac{P_0 a^2}{2} \left(1 - \frac{x}{a}\right) = P_0 \left(\frac{x^2}{2} - \frac{ax}{2} - \frac{a^2}{6}\right) = M_r(x)$$

$$Q = \frac{d}{dx} M(x) = P_0 \left(x - \frac{a}{2}\right)$$

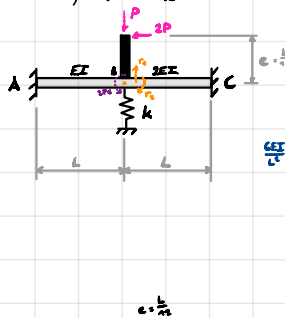
$$x=0: -\frac{P_0 a^2}{6} \quad x=2a: \left(\frac{4a^2}{2} - \frac{2a^2}{2} - \frac{a^2}{6}\right) = \frac{5P_0 a^2}{6}$$

$$x=0: Q = -\frac{P_0 a}{2} \quad x=2a: Q = \frac{3P_0 a}{2}$$

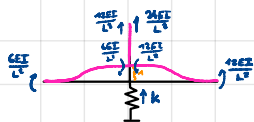
Problem 2) a) Find $\underline{K}\vec{r} = \vec{R}$

S1)

direct stiffness method:



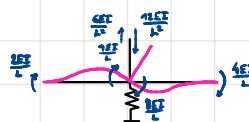
Case 1: $r_1 = 1$ $r_2 = 0$:



$$K_{11} = \frac{12EI}{L^3} + \frac{24EI}{L^3} + k = \frac{36EI}{L^3} + k$$

$$K_{21} = \frac{6EI}{L^2} - \frac{12EI}{L^2} = -\frac{6EI}{L^2}$$

Case 2: $r_1 = 0$ $r_2 = 1$:



$$K_{12} = \frac{6EI}{L^2} - \frac{12EI}{L^2} = -\frac{6EI}{L^2}$$

$$K_{22} = \frac{4EI}{L} + \frac{8EI}{L} = \frac{12EI}{L}$$

$$\underline{K} = \begin{bmatrix} \frac{36EI}{L^3} + k & -\frac{6EI}{L^2} \\ -\frac{6EI}{L^2} & \frac{12EI}{L} \end{bmatrix}$$

S2) $\underline{R} = \begin{bmatrix} -P \\ -2Pe \end{bmatrix} = \begin{bmatrix} -P \\ -\frac{PL}{6} \end{bmatrix}$

$$\underline{K}\vec{r} = \vec{R} \Rightarrow \begin{bmatrix} \frac{36EI}{L^3} + k & -\frac{6EI}{L^2} \\ -\frac{6EI}{L^2} & \frac{12EI}{L} \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} -P \\ -\frac{PL}{6} \end{bmatrix}$$

b) Find $\vec{r}$, when $k=0$

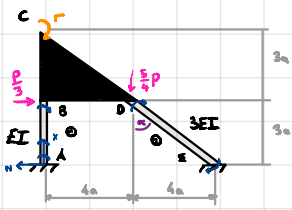
$$k=0 \Rightarrow \underline{K}_0 = \begin{bmatrix} \frac{36EI}{L^3} & -\frac{6EI}{L^2} \\ -\frac{6EI}{L^2} & \frac{12EI}{L} \end{bmatrix}$$

$$\det(\underline{K}_0) = \frac{36EI}{L^3} \cdot \frac{12EI}{L} - \frac{6EI}{L^2} \cdot \frac{6EI}{L^2} = \frac{432E^2I^2}{L^4} - \frac{36E^2I^2}{L^4} = \frac{396E^2I^2}{L^4}$$

$$\Rightarrow \underline{K}_0^{-1} = \frac{L^4}{396E^2I^2} \begin{bmatrix} \frac{12EI}{L} & \frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{36EI}{L^3} \end{bmatrix} = \frac{1}{396EI} \begin{bmatrix} 12L^3 & 6L^2 \\ 6L^2 & 36L \end{bmatrix}$$

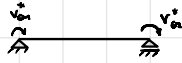
$$\vec{r} = \underline{K}_0^{-1} \vec{R} = \frac{1}{396EI} \begin{bmatrix} 12L^3 & 6L^2 \\ 6L^2 & 36L \end{bmatrix} \begin{bmatrix} -P \\ -\frac{PL}{6} \end{bmatrix} = \frac{1}{396EI} \begin{bmatrix} -13L^3P \\ -12L^2P \end{bmatrix} = \vec{r}$$

Problem 3) a) find $\hat{g}$



$$\underline{\underline{K}} = \sum_{i=1}^N (\underline{\underline{a}}^{(i)})^T \underline{\underline{k}}^{(i)} \underline{\underline{a}}^{(i)}$$

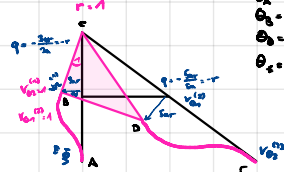
$$\alpha = \tan^{-1}\left(\frac{4a}{3a}\right) = \tan^{-1}\left(\frac{4}{3}\right)$$



$$\underline{\underline{S}} = \begin{bmatrix} S_{11} \\ S_{12} \\ S_{21} \\ S_{22} \end{bmatrix} = \frac{3EI}{L} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} V_{01} \\ V_{02} \end{bmatrix}$$

$$\begin{bmatrix} V_{01} \\ V_{02} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} r$$

$$\begin{bmatrix} V_{03} \\ V_{04} \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} r$$



$$\begin{aligned} \theta_1 &= 0 & S_{11} &= -\frac{3EI}{L} = -r \\ \theta_2 &= r & S_{12} &= -\frac{6EI}{L} = -r \\ \theta_3 &= r & S_{21} &= -\frac{6EI}{L} = -r \\ \theta_4 &= 0 & S_{22} &= -\frac{3EI}{L} = -r \end{aligned}$$

$$r^* = \theta^* = 0$$

$$\begin{aligned} V_{01} &= 0 - (-r) = r \\ V_{02} &= r - (-r) = 2r \\ V_{03} &= r - (-r) = 2r \\ V_{04} &= 0 - (-r) = r \end{aligned}$$

b) Find $\underline{\underline{K}}$

$$\underline{\underline{K}} = \sum_{i=1}^N \underline{\underline{a}}^{(i)T} \underline{\underline{k}}^{(i)} \underline{\underline{a}}^{(i)}$$

$$1) [1 \ 2] \cdot \frac{3EI}{L} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$2) [2 \ 1] \cdot \frac{6EI}{L} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{4EI}{L} + \frac{4EI}{L} \\ \frac{2EI}{L} + \frac{8EI}{L} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{8EI}{L} + \frac{20EI}{L} = \frac{28EI}{L} = \frac{28EI}{3a}$$

$$\begin{bmatrix} \frac{24EI}{L} + \frac{6EI}{L} \\ \frac{12EI}{L} + \frac{12EI}{L} \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \frac{60EI}{L} + \frac{24EI}{L} = \frac{84EI}{5a}$$

$$\underline{\underline{K}} = \frac{28EI}{3a} + \frac{84EI}{5a} = \frac{392EI}{15a} = \underline{\underline{K}}$$

c) Find $\hat{R}$

$$\hat{R} = \frac{5}{4}P \cdot 4a - \frac{1}{3}P \cdot 2a = 4Pa = \hat{R}$$

$$\underline{\underline{K}} \hat{r} = \hat{R} = \frac{392EI}{15a} r = 4Pa \Rightarrow r = \frac{60Pa^2}{392EI}$$

d) Find moment distribution

$$\underline{\underline{S}}^* = \underline{\underline{k}}^* \underline{\underline{a}}^* r$$

$$\underline{\underline{S}}_{AB}^* = \frac{3EI}{3a} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} r$$

$$= \begin{bmatrix} \frac{4EI}{3a} + \frac{4EI}{3a} \\ \frac{2EI}{3a} + \frac{8EI}{3a} \end{bmatrix} \frac{60Pa^2}{392EI} = \frac{60Pa}{392} \begin{bmatrix} \frac{8}{3} \\ \frac{10}{3} \end{bmatrix} = \frac{20Pa}{392} \begin{bmatrix} 8 \\ 10 \end{bmatrix} = \frac{10Pa}{98} \begin{bmatrix} 8 \\ 10 \end{bmatrix} = \begin{bmatrix} \frac{80Pa}{98} \\ \frac{100Pa}{98} \end{bmatrix} = \begin{bmatrix} \frac{20Pa}{49} \\ \frac{25Pa}{49} \end{bmatrix}$$

$$\underline{\underline{S}}_{DC}^* = \frac{6EI}{5a} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} r$$

$$= \frac{6EI}{5a} \begin{bmatrix} 5 \\ 4 \end{bmatrix} \frac{60Pa^2}{392EI} = \begin{bmatrix} \frac{480Pa}{490} \\ \frac{360Pa}{490} \end{bmatrix} = \begin{bmatrix} \frac{48Pa}{49} \\ \frac{36Pa}{49} \end{bmatrix}$$

